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Palaeomagnetism of the Permian rocks of Lodève (Hérault, France)

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Abstract

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The Lodève basin (43°41.5′N; 3°21′E) is a reference site for the determination of the Permian magnetic poles of the Western European Block.

Sedimentary formations (pelites) from the Lodève basin (43°41.5′N latitude, 3°21′E longitude), of Permian (Saxonian and Autunian) age, have been studied by applying thermal and alternating fields, and thermal followed by alternating field demagnetization. The principal Saxonian magnetization has a mean, derived from 93 specimens giving unit weight to the six sites, of $D = -160.6°$, $I = -11.8°$, $\alpha_{95} = 1.5°$, and a corresponding north paleopole at 48.6° N, 153.5° E, $K = 2096$, $A_{95} = 1.2°$. The mean direction of the principal Autunian magnetization from 65 specimens, giving unit weight to each of the eight sites, is $D = -169.7°$, $I = 6.4°$, $\alpha_{95} = 3.0°$. This direction corresponds to a north paleopole at 42.2° N, 169.4° E, $K = 517$, $A_{95} = 2.2°$. These Autunian and Saxonian poles are very close, respectively, to intervals 270–290 Ma and 240–260 Ma of the European polar wander path, close to the Permian poles derived from the Lodève basin by Kruseman (1962) (Saxonian and Autunian poles are situated, respectively, at 48° N, 164° E and 44.5° N, 178° E).

Introduction

Examination of the Permian magnetic poles which have allowed the definition of the European apparent polar-wander (APW) curve (Irving and Irving, 1982; Westphal et al., 1986), shows that the relatively few poles established, for western Europe, by early work, are defined after simple alternating-field demagnetization. It appeared useful therefore to study the reference sites utilizing the recent progress in magnetic analysis, i.e. thermal treatment and also thermal treatment associated with (followed by) alternating-field demagnetization. The analysis of the Lodève series is particularly interesting because of recent palaeontological and stratigraphic studies, and the exceptional sampling conditions offered by the mining developments (uranium-bearing mineralizations) of the COGEMA company.

The first paleomagnetic study of the Lodève basin was made by Kruseman (1962). This author had analysed the natural remanent magnetization of 50 specimens of Autunian rocks and of 89 specimens of Saxonian rocks, by alternating-field demagnetization. Kruseman had difficulty separating the different components of magnetizations carried by many specimens.

Geological setting

The Permian Lodève basin is situated south of the Massif Central (43°41.5′N; 3°21′E; Fig. 1a). This basin is limited to the north by the Causses, and separated by faults from the Montagne Noire

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to the west, and from the Languedoc basin to the southeast.

The Lodève continental Permian sediments lie on Cambrian rocks and their thickness locally exceeds 2000 m. Two groups of sediments are classically distinguished in this series. The first is classically considered as Autunian, and begins with a basic conglomerate, followed by siltites and argillites. Grey sediments predominate at the base (Usclas and Loiras sets) and red ones predominate in the top half (Mass d'Alary and Viala sets). The second, attributed to the Saxonian, lying unconformably on its substratum with a basal conglomerate, is essentially formed by red pelites and sands.

Uranium impregnates some levels and is found in particular facies (bituminous impregnations) or along faults. Prospectors have found several cineritic levels in the base of the Permian series; these levels are usually denoted with roman numerals. They facilitate lateral correlations and have been employed for detailed stratigraphic and structural analyses of the basin (Laversanne, 1976).

Ellenberger (1983) defined an accurate ichnological zonation in the Permian, on the basis of numerous vertebrate tracks, and compared the Lodève Permian series to their German and African equivalents. Recent studies (Conrad et al., 1986; Odin et al., 1986) seem to indicate that "Autunian" cinerites V and XI are of Saxonian age. The first study was made using the K–Ar method of isotope dating on the feldspars of four Autunian cinerites. The second study takes spore and pollen associations into account. Nevertheless, the evidence of radiometric resetting ages produced by these authors has led us to keep the classical stratigraphic levels.

The basin corresponds to a zone of subsidence, the monoclinal filling of which dips to the south (10°–20°). It was progressively deformed and faulted synchronously with a meridian oriented extension and synsedimentary subsidence (Horrenberger and Ruhland, 1981).

Sampling and measurement procedures

Samples were collected in 1984 at 17 sites (Fig. 1), as follows:

(a) The lower part of the Usclas set: site numbers 5 and 12–15. Sites 12–15 are situated in the mine (level −700 m, near the prospectors' layer number 35); site 5 is situated in the Usclas ravine (Laversanne's section, 1976; layers 34–35).

(b) The upper part of the Usclas set: sites 16 and 17 (COGEMA quarry; layers 21–23).

(c) The upper part of the Mas d'Alary set: sites 1–3 (red Autunian levels, but of Saxonian age from palynological data; National road number 9, crossing the Mas d'Alary road).

(d) The lower part of the classical Saxonian formation: sites 6–11 (National road number 9, Cartels village).

Sampling was made for the most part using plaster of Paris caps for orientation (Thellier, 1938), with the geographical northern azimuth being generally (when it was possible) controlled with a sun sighting. Some samples were directly drilled (sites 1–5). Samples from site 4 were unusable after transportation.

In the laboratory, two to five specimens were drilled from each sample, so as to enable systematic magnetic (alternating field) and thermal cleaning. The two techniques were employed successively (thermal demagnetization, and then alternating-field demagnetization) on each grey specimen to eliminate the overprints of different components of the magnetization. Daly (1981) has detailed the advantages of the combined procedure. Indeed, the grey rocks undergo mineralogical transformations at temperatures higher than 350°C, producing parasitic thermoremanence which inhibits the determination of the principal magnetization.

The remanent magnetization carried by the specimens was measured using principally a JR − 4 spinner magnetometer (Geophysica, Brno) and a translation inductometer (Daly, 1967). Two devices, recently constructed in the Saint Maur laboratory were utilized for specimen demagnetization. The first is a cylindrical furnace with a fast heating level, reached in 15 min and maintained for 45 min, and in which the field is cancelled by four layers of Mu-metal, which are demagnetized before each treatment. The specimens are cooled for one hour by forced air at the heating site. The second apparatus, used for alternating field de-

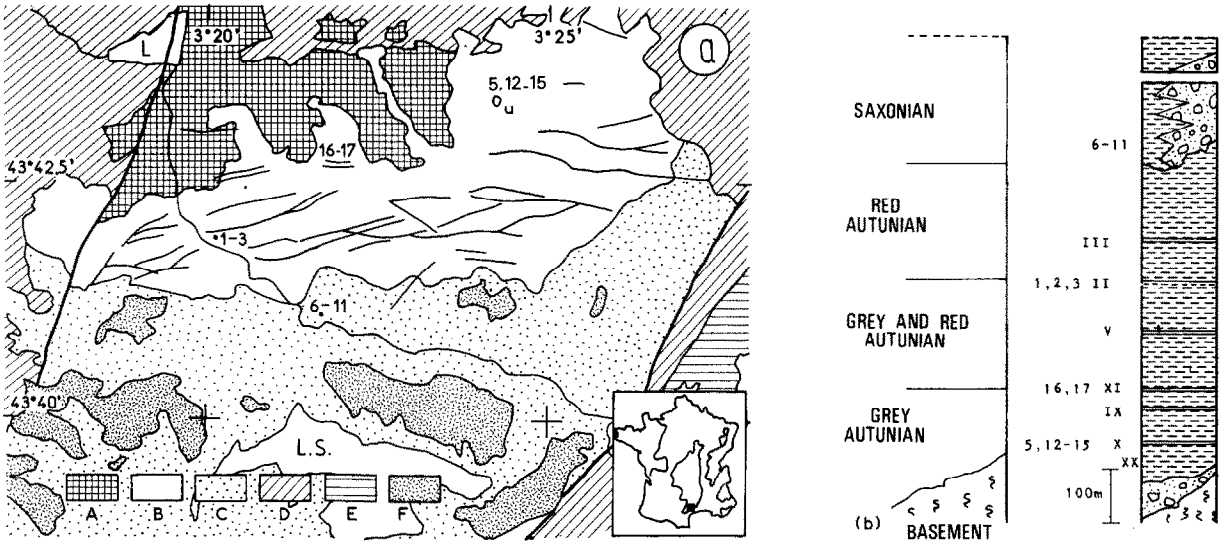


Fig. 1. a. Geological sketch map of the Lodève basin and sampling site locations: *A*—basement; *B*—Autunian; *C*— Saxonian; *D*—Secondary; *E*—Tertiary; *F*—Quarternary basalts; *L.S.*—Salagou lake; *L*—Lodève; *1–17*—sampling sites. b. Synthetic log of the Lodève Permian sediments: roman numerals, cinerites; Arabic numerals denote sampling sites.

magnetization, automatically cancels the parasite anhysteretic effect (Le Goff, 1985) up to 280 mT.

Treatment steps were varied according to the nature of the magnetization: when the magnetization moments were small, the alternating-field analysis was made at 2.5 mT intervals, and when the moments were relatively high, the intervals were 5 mT. The thermal demagnetization was conducted with steps of 100 °C up to 500 °C. For higher temperatures, the intervals were reduced (50 °C and 25 °C), to permit a more detailed study of the magnetic record evolution.

Data analysis was carried out using classical methods: (1) analysis of the remaining magnetization after each treatment (remaining vector); (2) analysis of the magnetization lost at each treatment (difference vector); and (3) the presentation of data on equal-area projection diagrams and on orthogonal-vector plots. Mean directions of the stable magnetizations were computed using Fisher's method (1953). When several components are superposed, principal component analysis (Kirschvink, 1980) was used to estimate the different directions of the components of magnetization or the plane containing them.

Magnetic viscosity tests were made to estimate the potential for acquisition of a viscous remanent magnetization (VRM) of a representative set of the specimens before any treatment. The viscosity index is expressed as the ratio dM/M, M and dM being, respectively, the magnetic moment vector measured after a prolonged storage in a zero magnetic field, and its variation after a stay of two weeks in the terrestrial field. For the present study, the viscosity index varies between 5% and 50% (mean value = 19%). Thus, to avoid the disturbing influence of VRM (Biquand and Prévot, 1970) during treatment, the specimens were systematically kept in a zero-field environment.

Magnetic behaviour

In all, 158 specimens were analysed. Two groups of different magnetic intensity are distinguished.

The first is formed by red Autunian and Saxonian rocks, which have relatively high magnetic intensities with values between 339×10^{-4} and 40×10^{-4} A m^{-1}, and a mean of 144×10^{-4} A m^{-1}. The second is formed by the grey rocks; their magnetization intensities are much smaller than those of the first group, with values between 53×10^{-4} and 10^{-4} A m^{-1}, and a mean of 16×10^{-4} A m^{-1}. The specimens from sites 13 and 15 (cinerite X) carry very small magnetization inten-

sities (beyond the limit of the measurement device's sensitivity).

The natural remanent magnetizations were measured after a long period in a zero-magnetic field, to eliminate the principal part of the VRM acquired *in situ* or after sampling. Two different groups of natural remanent magnetization directions, Autunian and Saxonian, have been found (Fig. 2).

The Autunian magnetization directions which are plotted in equal-area projection are distributed in a subvertical plane trending NNE–SSW (Fig. 2a). The pole of the least-squares plane fitted to these data lies at $D = 119°$ and $I = 1°$. This distribution and the mean direction of the magnetizations ($D = 174°$, $I = 38°$) suggest the presence of a second component, probably related to the actual Earth's magnetic field, adding to the inverse and low inclinations Permian magnetization expected for the site. For the quasi-totality of the specimens, the magnetization directions of the Saxonian rocks have a SSW azimuth and a low inclination (Fig. 2b). The mean direction ($D = -159°$, $I = -9°$) of these magnetizations is very close to the Saxonian direction expected for the site (magnetization with a low inclination). These rocks thus have conserved, at least in part, their original magnetization.

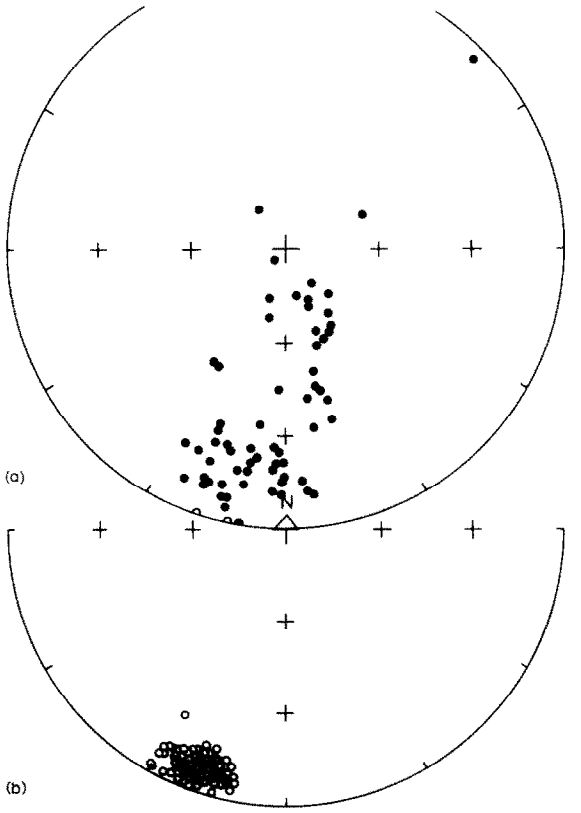


Fig. 2. Natural remanent magnetizations pointed after dip correction of the Autunian (a) and of the Saxonian (b) specimens, before magnetic analysis. Data are plotted on the whole sphere of an equal-area projection. Circle shows negative inclination; point shows positive inclination.

The thermal and alternating field analyses show several major trends, apparently associated with rock composition and also with the intensity of the natural remanent magnetization. The mineralogy was investigated by saturating the remanent magnetization by magnetic field and thermal demagnetization of the saturated magnetization (Lowrie and Heller, 1982). The red rock's magnetic carrier is haematite (saturation field > 1.5 T; thermal demagnetization at 670 °C) and that of the grey rocks is magnetite (saturation field included between 0.3 and 0.4 T; thermal demagnetization at 580 °C)

The thermal analysis of the haematite bearing specimens was very efficient in separating the different components. On the other hand, thermal treatment associated with alternating field demagnetization was necessary to separate components in the grey specimens, which carried a small magnetization intensity or were altered. Three types of magnetic behaviour were distinguished:

(a) The remaining vectors and the difference vectors reach a steady direction for the first stages of the treatment. The most frequent case is presented in Fig. 3, in which only one Permian component of reverse polarity exists. The magnetization is carried by grains with high blocking temperatures. However, the existence of a second component, which is destroyed at low temperatures, is generally shown by the first three difference vectors. The direction of this component may be connected with the present Earth's field. This has been found to be the case in several red Autunian and Saxonian rocks.

(b) The second type shows two components,the blocking-temperature spectra of which are clearly separate. An equal-area projection (Fig. 4) shows

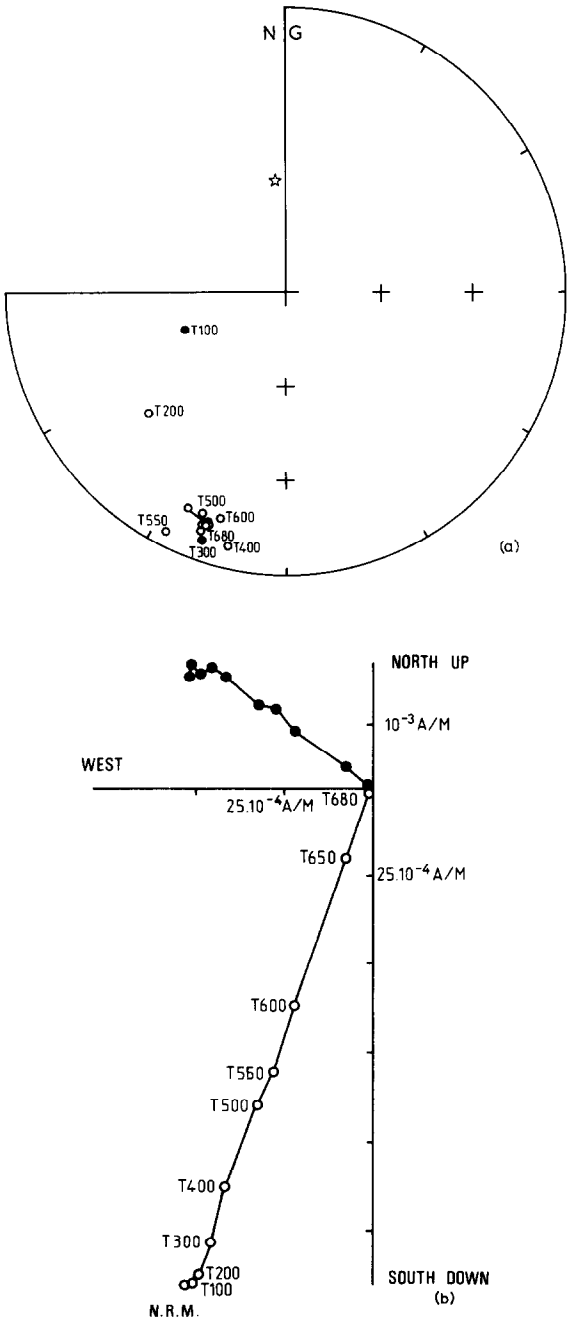


Fig. 3. On an equal-area projection, a single reverse polarity component is defined by close grouping of remaining vectors (symbols joined by a line) and difference vectors (open and solid symbols are used respectively for negative and positive inclination in Fig. 3a. The latter are identified by their code treatment: T—temperature (°C). On the Zijderveld diagram (orthogonal plot, Fig. 3b), open circles correspond to projections of remaining vectors on a horizontal plane. Solid circles are used for the vertical remaining vectors. Star: present direction of the Earth's field.

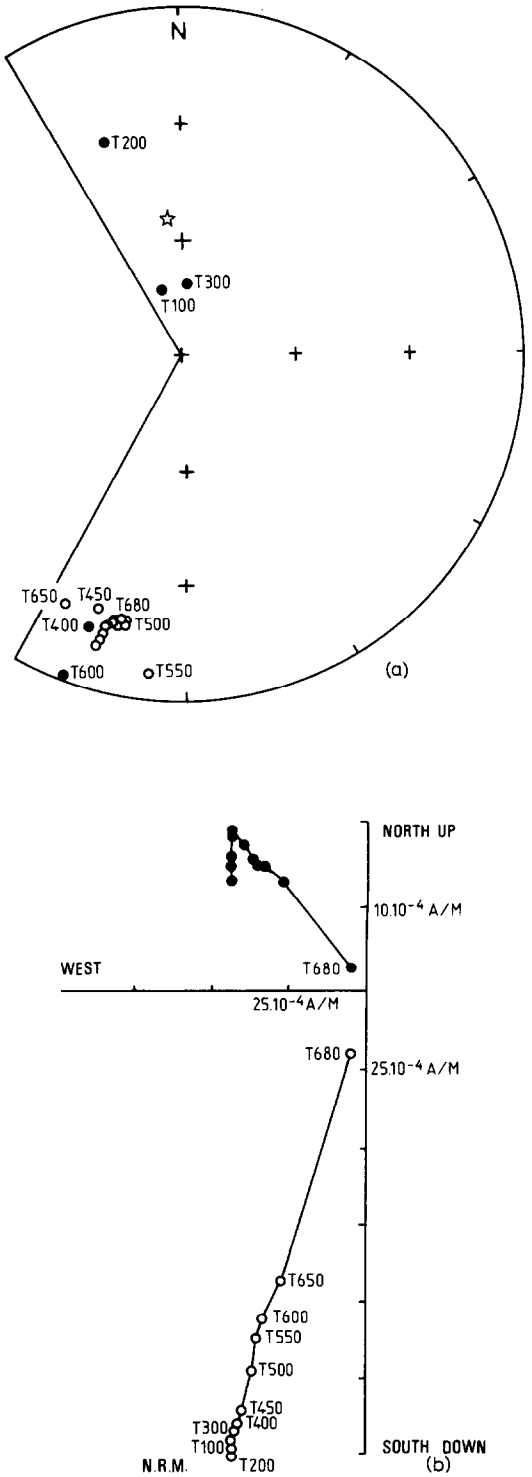


Fig. 4. Equal-area plot showing a dominant inverse polarity component defined by close-grouped remaining vectors and difference vectors to the SSW; the presence of the actual viscous component is indicated by the difference vector T100, T200 and T300. See Fig. 3 for key.

a viscous component (present field), carried by grains with a blocking temperature less than 300 °C; it is well defined by the first two difference vectors. Only a reverse Saxonian component remains at temperatures higher than 300 °C. This behaviour is typical of about 30% of the red Autunian and Saxonian specimens.

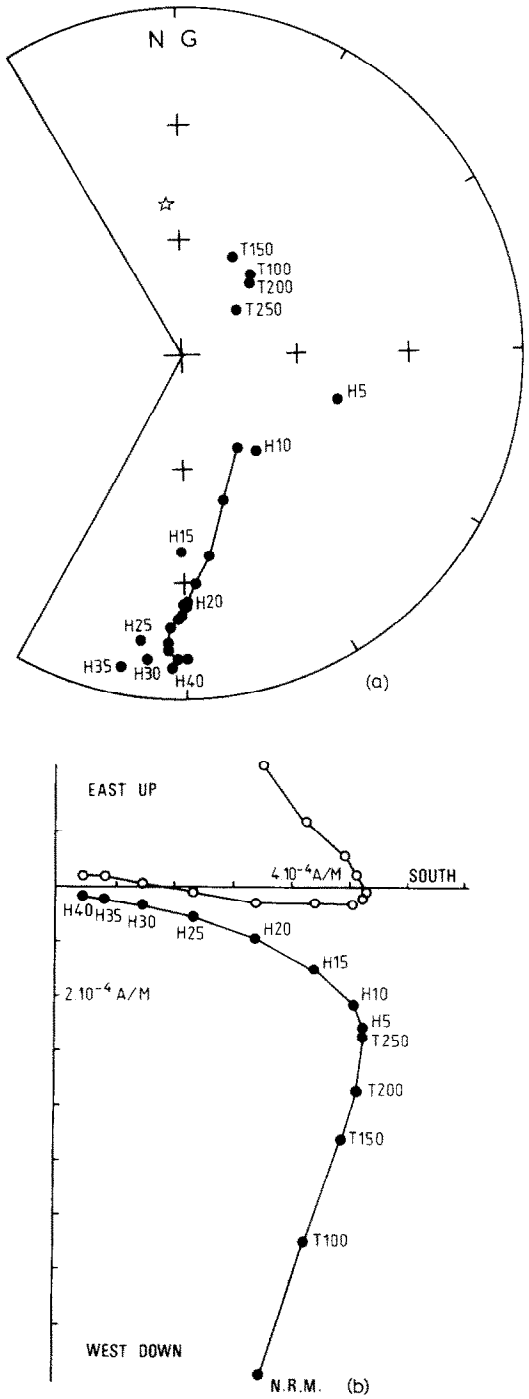


Fig. 5. Equal-area plot showing an evolution of the remaining vectors towards the direction of a reverse polarity component. This latter component is defined by the last difference vectors (H30, H35 and H40 in mT). See Fig. 3 for key.

(c) The last type is found only in Autunian specimens of grey colour. In Fig. 5, an equal-area projection shows a viscous component, defined by the difference vectors from thermal treatment, due to the present field; it is isolated from the Autunian component which appears in the final difference vectors (alternating-field treatments).

Palaeomagnetic implications

The magnetic analysis has allowed the isolation of a remanent magnetization with Permian direction and reverse polarity in both Autunian and Saxonian rocks. This component is well defined in the red Autunian (sites 1, 2 and 3) and in the Saxonian rocks; but a detailed analysis also indicates its presence in the grey Autunian rocks. Its inclination is negative (about $-11°$) for the Saxonian rocks and slightly positive (about $7°$) for the Autunian rocks. On this primary component is superposed a VRM, acquired in the present Earth's field. The relative importance of this viscous component is particularly high in the grey Autunian rocks.

The study results are summarized site by site in Table 1. Taking into account the structural conditions, the results are given before and after dip corrections.

It must be noted that sites 1–3, which are in the classical Autunian (Mas d'Alary group) but from which the cinerites (V and XI) contain a sporo-polyclinical Saxonian association, seem to be placed in the Autunian from a palaeomagnetic point of view.

Fig. 6a shows a notable amelioration of the directional grouping obtained after analysis of the 65 Autunian specimens. The radius of the confidence cone is reduced from 6.7° to 1.8°. The mean direction, computed giving equal weights to each specimen, is defined by $D = -169.9°$ and $I = 7°$ (after dip correction). This direction does not differ greatly from that obtained by giving equal weight to each site ($D = -169.7°$, $I = 6.4°$). The grouping of directions from 93 Saxonian

TABLE 1

Summary of results obtained after magnetic analysis *

	Sites	N	D_m (°)	I_m (°)	α_{95} (°)	λ (°N)	φ (°E)	K	A_{95} (°)	A_{63} (°)	(P, A)
Saxonian red	6	19	−160.6	−11.3	2.1	48.4	153.6				(12, 180)
sandstones	7	12	−160.5	−13.2	2.0	49.2	152.9				(12, 180)
	8	13	−162.9	−10.7	2.2	48.8	157.0				(12, 180)
	9	21	−160.1	−9.4	2.1	47.3	153.4				(12, 180)
	10	17	−161.5	−12.6	2.5	49.3	154.5				(12, 180)
	11	11	−158.1	−13.3	3.5	48.3	149.5				(12, 180)
Autunian red	1	9	−172.5	5.6	5.7	43.1	173.3				(12, 180)
sandstones	2	7	−170.0	−0.5	3.6	45.6	169.1				(12, 180)
	3	5	−167.3	5.4	6.7	42.2	166.2				(12, 180)
Autunian grey sandstones	16	14	−172.5	8.7	3.1	41.5	173.4				(19, 130)
(level XI to XII)	17	9	−168.5	10.1	3.2	40.1	168.3				(19, 130)
Autunian grey sandstones	5	6	−171.2	6.3	4.7	42.5	171.4				(12, 180)
(level X to XX)	12	8	−163.4	6.5	3.4	40.7	161.3				(10, 173)
	14	7	−171.5	8.9	5.5	41.2	172.1				(10, 173)
All Saxonian samples											
B.D.C.		93	−161.1	−0.2	1.0	43.3	157.0				
A.D.C.		93	−160.7	−11.5	1.0	48.5	153.7				
All Saxonian sites											
B.D.C.		6	−161.1	−0.3	1.6	43.3	156.9	2192	1.2	0.7	
A.D.C.		6	−160.8	−11.8	1.5	48.6	153.5	2096	1.2	0.7	
All Autunian samples											
B.D.C.		65	−168.1	17.4	1.7	36.3	168.8				
A.D.C.		65	−169.9	7.0	1.7	41.9	169.8				
All Autunian sites											
B.D.C.		8	−168.8	16.9	2.5	36.5	168.9	479	2.3	1.3	
A.D.C.		8	−169.7	6.4	3.0	42.2	169.4	517	2.2	1.3	
All samples											
B.D.C.		158	−163.9	7.0	1.6	40.6	162.0				
A.D.C.		158	−164.5	−3.9	1.8	46.0	160.8				
All sites											
B.D.C.		14	−165.1	9.6	4.7	39.6	164.0	155	3.0	1.7	
A.D.C.		14	−165.8	−1.4	5.0	45.2	163.0	127	3.3	1.9	

* N is the number of specimens for each site, K the precision parameter, Dm, Im, λ, φ, α_{95}, A_{95} and A_{63} are respectively, in degrees, the declination and inclination of the magnetization, latitude and longitude of paleomagnetic poles, the radius of the circle of confidence, and errors of the paleopoles at 0.95 and 0.63 probability, giving unit weight to each of six Saxonian and eight Autunian site poles. B.D.C. and A.D.C., before and after dip correction. Geological stratigraphic correction; dip P bearing A.

specimens is better than that before analysis, as shown in Fig. 6b. The mean direction of the Saxonian magnetization is defined by $D = -161°$ and $I = -11.5°$ (equal weight given to each specimen).

The mean pole computed after dip corrections (Table 1) from the six Autunian sites is located at 42.2°N and 169.4°E (with $K = 517.3$ and $A_{95} = 2.2°$); the mean pole computed from the eight Saxonian sites is located at 48.6°N and 153.5°E (with $K = 2096$ and $A_{95} = 1.2°$).

These results are close to those of Kruseman

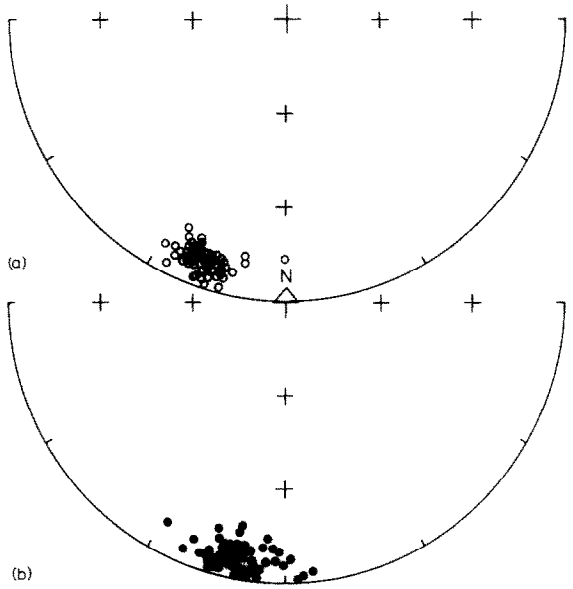


Fig. 6. Magnetization directions after dip correction of the 65 Autunian (a) and of the 93 Saxonian (b) specimens, after analysis and dip corrections. Circle shows negative inclination; point shows positive inclination. Data are plotted on the whole sphere of an equal-area projection.

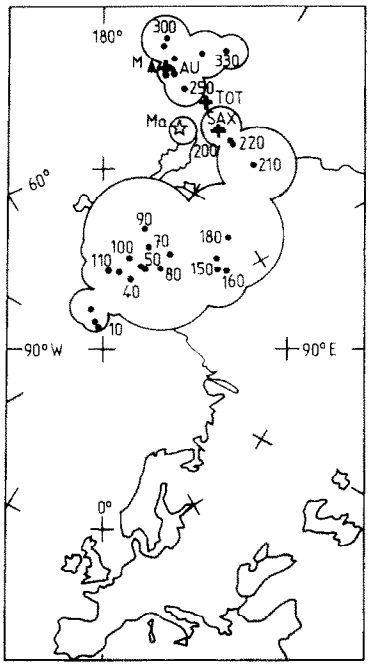


Fig. 7. The apparent mean polarwander path for northern Eurasia (data from Irving and Irving, 1982, using a 30 Ma sliding window). Triangle (*M*), Moissey (French Jura) pole position; star (*Ma*), Maures (southeast France) pole position; crosses, Lodève Autunian (*AU*) and Saxonian (*SA*) pole positions.

(Autunian 44.5° N, 178° E; Saxonian 48° N, 164° E); nevertheless, the successive applications of thermal cleaning and alternating-field demagnetization have allowed a good definition of the fossil magnetization carried by the grey rocks. This affirmation is justified by the position of the pole derived from the Autunian red rocks ($\lambda = 44°$ N, $\phi = 170°$ E), which is very close to that of the pole derived from the Autunian grey rocks ($\lambda = 41°$ N, $\phi = 169°$ E). It is clear that the application of more modern techniques, such as those used in the present study, allows a more detailed and precise analysis of these rock types. The results can be associated with precise biostratigraphic data, and so a finer elaboration of the magnetostratigraphy is now possible.

The pole positions are shown on Fig. 7, relative to the APW path for northern Eurasia, determined by Irving and Irving (1982). This figure also shows the poles recently obtained for the Permian sediments of the Maures (Merabet and Daly, 1986) and for the Permian ignimbrites of the Jura Mountains (Thompson et al., 1986). While the Maures pole lies at a greater distance from the curve than the standard angular deviation (which might reveal a small rotation of Maures), the two Lodève poles are correctly placed bearing in mind their corresponding age intervals. The rotational movements which could have affected the Lodève region since the Permian, if they have not been compensated by reverse movements, were not of sufficient importance to be detected by palaeomagnetic analysis.

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